Critical Memory and Information Congestion in Stateful Pricing Learners

Anonymous Authors

Abstract

A stateful router can make a profitable price cut hard to learn without making it unprofitable. We separate three boundaries. A rational threshold determines whether a persistent cut is valuable. A finite-time threshold follows because a learner cutting with conditional probability at most q sees a fresh length-M path within T steps with probability at most Tq^M. Across n providers, the expected responsive set is therefore at most $T \sum_i q_i^{M_i}$. Combining this bound with signals of effective rank $r_n \asymp n^\beta$ yields a conditional efficient adaptive set $k^* \asymp n^{(1+\alpha\beta)/(1+\alpha)}$ and an explicit memory level that caps finite-horizon exposure near k^*. A path-equivalent commitment option reduces proposal time to linear while preserving primitive paths and optimal value. In a frozen MDP it changes correct Q-learning from 1/20 to 19/20 seeds at $M = 7$, but overcuts beyond the rational boundary. A fixed-n bandit experiment changes action correlation without changing allocation. Finally, 3.18 million public endpoint observations show a large GLM-5.2 shadow-share surface but reject the proposed GLM-minority/non-GLM-linear empirical split. The contribution is a finite-time learning theory with falsifiable transport tests, not a claim of live-provider collusion or router memory.

1 Introduction

Independent pricing algorithms interact through market feedback even without communication. Existing results emphasize reward signal quality, parallel experimentation, and the scoring rule faced by each learner (Hansen et al., 2021; Calvano et al., 2020). A different friction appears when the platform remembers actions. A price cut can pay poorly today yet become profitable only after enough consecutive cuts erase a ranking penalty. No isolated experiment reveals that long-run return. At the same time, many providers may respond to the same low-dimensional signal. Temporal credit and cross-sectional signal rank jointly determine how many sellers can learn to adapt within a market horizon.

AI inference routing is a useful application because a router aggregates token demand and selects among providers serving the same open-weight model. Public menus expose contemporaneous quotes; private eligibility, non-price scores, fallback, and capacity remain hidden. Current data therefore reveal the mechanical market-share surface of a declared price rule, not the live memory state. We use the public panel as a property test: it can falsify proposed cross-sectional regimes, while the controlled environment isolates temporal credit. No paid probe currently randomizes a memory rule.

This distinction matters for both algorithmic-collusion and mechanism-design claims. A high learned price can arise because high pricing is an equilibrium, because experiments are statistically coupled, or because a learner never discovers a profitable delayed path. These mechanisms demand different interventions. More exploration addresses the last mechanism only at an exponential cost; changing the action interface can solve it but may induce overcommitment.

Our contributions are sixfold.

1. We derive an exact *rational* boundary M^* for persistent cutting under a finite-memory penalty. It separates environments where the long-run low price is optimal from those where the high price is optimal.
2. We derive a distinct *learning* boundary. After any stopping time σ, an adaptive policy whose cut probability is at most q has conditional probability at most Tq^M of producing a fresh length-M path within T steps. Combining this bound with ordinary reward estimation separates late temporal credit from reward SNR.
3. We lift the path bound to a market of n providers. It bounds the finite-horizon responsive count and, under inverse-power routing, the local total-variation reallocation caused by newly learned quote changes.
4. We combine this learning bound with an information-congestion planner. If provider signals have effective rank $r_n \asymp n^\beta$, the efficient adaptive set scales as $n^{(1+\alpha\beta)/(1+\alpha)}$. A logarithmic memory rule can implement its expected exposure cap, but only when persistent adaptation remains on the rational side of the first boundary.
5. We show that a length-$(M+1)$ commitment option preserves the optimal value and feasible primitive paths while replacing exponential path-proposal time by a linear term. This is an interface result, not a change in provider payoffs.
6. We test the mechanism in a frozen, prespecified finite MDP. The option closes the local gap at intermediate memory and becomes harmful beyond the rational boundary. A frozen trace replay separates early dis-

covery from late credit starvation. State observability, cross-market severity, and a conventional multi-step return each fail prespecified tests.

7. We add two transport falsifications. A fixed-n bandit intervention moves simulated action correlation but not active share. In public data, GLM-5.2 has three active undercutters in a 25-provider menu and a large inverse-price shadow transfer, yet its effective-rank slope is linear-compatible; 90% of non-GLM markets have no active undercutter. Thus the attractive GLM-$o(n)$/non-GLM-$\Omega(n)$ contrast is not a finding.

2 Stateful pricing game and economic boundary

A seller repeatedly chooses a high quote H or low quote L. The buyer's declared allocation rule carries a binary penalty state. A low quote remains penalized until the previous M quotes are also low; after that run, the penalty clears. The state can represent a prominence rule, a reserve-ranking flag, or any other disclosed history-dependent procurement policy. We do not assume that a particular live router implements it.

Let u_H be seller profit at H, $u_{\theta L}$ penalized profit at L, and u_L unpenalized profit at L. These quantities may be generated by any demand or allocation system. The theory requires only the economically interesting ordering is

$$u_L > u_H > u_{\theta L}. \tag{1}$$

ordering: cutting loses while the flag is active but wins after it clears.

Theorem 1 (Rational memory boundary)**.** *From an all-high history, an infinite-horizon optimal policy either stays high forever or cuts immediately and stays low. For discount $\gamma \in (0,1)$, cutting is strictly optimal iff*

$$\gamma^M > \frac{u_H - u_{\theta L}}{u_L - u_{\theta L}}. \tag{2}$$

The continuous boundary is $M^* = \log[(u_H - u_{\theta L})/(u_L - u_{\theta L})]/\log\gamma$.

In the frozen numerical profile, $(u_H, u_{\theta L}, u_L) = (0.1067535, 0.0351280, 0.1501829)$ and $\gamma = 0.95$, giving $M^* = 9.2402$. Exact dynamic programming cuts for integer $M \le 9$ and stays high for $M \ge 10$. This boundary is derived before examining learning outcomes.

3 Critical memory for finite-time learning

Let $A_t \in \{H, L\}$. Fix any stopping time σ representing a stage at which downstream reward or value information has become usable, and define $\tau_M(\sigma)$ as the first $t \ge \sigma + M$ for which the M actions after σ ending at t are all low. This definition is intentionally conditional: an early visit can reveal the terminal reward without supplying a later one-step learner enough fresh support to propagate that value upstream. We allow full history dependence and require only

$$\mathbb{P}(A_t = L \mid \mathcal{F}_{t-1},\, t > \sigma,\, \tau_M(\sigma) \ge t) \le q. \tag{3}$$

Theorem 2 (Exponential fresh-path barrier)**.** *Under equation (3), for every horizon T and almost every realized $\mathcal{F}_\sigma$,*

$$\mathbb{P}(\tau_M(\sigma) \le \sigma + T \mid \mathcal{F}_\sigma) \le (T - M + 1)_+ q^M \le T q^M. \tag{4}$$

Consequently, attaining conditional fresh-path probability at least $1-\delta$ requires

$$T \ge (1-\delta) q^{-M}. \tag{5}$$

If post-σ actions are iid Bernoulli(q), the exact expected additional time is

$$\mathbb{E}[\tau_M(\sigma) - \sigma \mid \mathcal{F}_\sigma] = \frac{1 - q^M}{(1-q)q^M}. \tag{6}$$

Define the finite-horizon critical memory

$$M_c(T, q, \delta) = \left\lfloor \frac{\log[T/(1-\delta)]}{\log(1/q)} \right\rfloor. \tag{7}$$

For $M \ge M_c + 1$, the target fresh-path probability is ruled out after any chosen readiness time σ. Thus memory is a computational control parameter for learners that need fresh downstream support: increasing T by a factor of ten buys only $1/\log_{10}(1/q)$ additional memory periods.

This clarifies the relationship to signal-to-noise results. Suppose reward differences are sub-Gaussian with gap Δ and proxy variance σ^2. Then $O(\mathrm{SNR}^{-1}\log(1/\delta))$ samples, where $\mathrm{SNR} = \Delta^2/\sigma^2$, suffice to identify the better downstream action. If upstream credit requires a fresh path after that readiness stage, equation (5) still applies.

Corollary 3 (Temporal–statistical crossover)**.** *For a fixed-confidence ordered-credit learner, total difficulty contains a fresh-support term q^{-M} and a statistical term $\mathrm{SNR}^{-1}\log(1/\delta)$. Temporal discovery dominates once*

$$q^{-M} \gg \mathrm{SNR}^{-1}\log(1/\delta). \tag{8}$$

Higher reward SNR cannot remove this conditional gap; it moves the readiness time σ but does not change post-σ path probability.

The result complements Hansen–Misra–Pai: independent experiments may couple through the final pricing score even with no explicit memory (Hansen et al., 2021). Our theorem does not contradict asymptotic learnability. It supplies a finite-time rate: once market memory requires a run rather than an isolated experiment, the time needed to re-expose the relevant score after downstream learning can be exponential in M. This conditional statement is not, by itself, an explanation of the Q-learning result; the trace audit below tests whether late support is actually absent.

4 From memory to market-share response

The preceding result concerns one learner. We now hold the calendar horizon fixed and ask how many providers can become newly responsive. Provider i has memory M_i and satisfies equation (3) with cap q_i after its own readiness stopping time. Let $I_i(T)$ indicate that it completes a fresh qualifying path within the next T periods and let $K_T = \sum_{i=1}^n I_i(T)$.

Proposition 4 (Finite-horizon responsive set)**.** *Without independence across providers,*

$$\mathbb{E}[K_T \mid \mathcal{F}_\sigma] \le \sum_{i=1}^n \min\{1, Tq_i^{M_i}\},$$
$$\mathbb{P}(K_T \ge k \mid \mathcal{F}_\sigma) \le \frac{\sum_i \min\{1, Tq_i^{M_i}\}}{k}. \tag{9}$$

The bound is agnostic about whether provider signals are independent, correlated, or identical. It says that memory can make the realized adaptive set sparse even when each provider would benefit from a sufficiently persistent change.

To connect this count to allocation, consider the declared inverse-power rule $s_i(p) = p_i^{-a}/\sum_j p_j^{-a}$. Suppose a newly responsive provider changes log price locally by d_i, with $|d_i| \le \Delta$, and a nonresponsive provider does not move. Let $Ds(p)[d]$ denote the first-order share change.

Proposition 5 (Local routed-share response)**.** *Conditional on the baseline menu,*

$$\mathbb{E}\left[\frac{1}{2}\|Ds(p)[d]\|_1\right] \le a\Delta \sum_{i=1}^n s_i(1-s_i)\min\{1, Tq_i^{M_i}\}. \tag{10}$$

Thus the mechanical own-price elasticity $a(1-s_i)$ and the probability of learning a fresh response enter multiplicatively. The bound is local and does not assert that the public rule is the realized router.

The number of responsive providers need not equal the efficient number. Let Σ_n be the covariance of the signals used by potentially adaptive providers and define effective rank

$$r_n = \frac{(\operatorname{tr}\Sigma_n)^2}{\operatorname{tr}(\Sigma_n^2)} \in [1, n]. \tag{11}$$

Consider the conditional planner objective

$$V_n(k) = \frac{bk}{n} - c\left(\frac{k}{n}\right)^2\left(\frac{k}{r_n}\right)^\alpha, \qquad b, c, \alpha > 0. \tag{12}$$

The benefit is the local allocation opportunity; the loss is a declared reduced form for common-mode overreaction. It is not implied by equation (10).

Theorem 6 (Information-congestion boundary)**.** *The unique continuous interior maximizer of equation* (12)*, before clipping to* $[0, n]$*, is*

$$k_n^\circ = \left(\frac{bnr_n^\alpha}{c(2+\alpha)}\right)^{1/(1+\alpha)}. \tag{13}$$

The integer optimum is adjacent to k_n°. If $r_n \asymp n^\beta$, then

$$k_n^* = \Theta\left(n^{(1+\alpha\beta)/(1+\alpha)}\right). \tag{14}$$

Hence $k_n^* = o(n)$ for $\beta < 1$ and $k_n^* = \Theta(n)$ for $\beta = 1$.

Corollary 7 (Memory required for an exposure cap)**.** *If $M_i = M$ and $q_i = q$, then any*

$$M \ge \frac{\log(Tn/k_n^*)}{\log(1/q)} \tag{15}$$

ensures $\mathbb{E}[K_T] \le k_n^*$ whenever the right-hand side is positive. Writing $k_n^* = \Theta(n^\zeta)$, the required memory grows as $[(1-\zeta)\log n + \log T + O(1)]/\log(1/q)$. A sublinear efficient adaptive set therefore requires logarithmically increasing temporal selectivity under this homogeneous implementation; a linear set does not.

This corollary is a mechanism identity, not a policy recommendation. Memory suppresses response by making evidence difficult to realize; it may exclude the wrong providers, and it can push a profitable cut past the rational boundary. A router should prefer direct covariance-aware exposure caps when signal groups are observable. The memory formula is useful because it makes the hidden cost of history-dependent scoring explicit.

5 A path-equivalent intervention

Augment the primitive MDP with a `commit-low` option that executes $M+1$ existing low actions. Both primitive actions remain available.

Theorem 8 (Value preservation and linear discovery)**.** *The augmented semi-Markov MDP has exactly the same optimal value as the primitive MDP at every state. It creates no new primitive price path. If the option is selected independently with probability q_o at each decision epoch while non-option decisions consume one transition, its completion time has expectation*

$$M + q_o^{-1}. \tag{16}$$

The option changes representation, not incentives. Any augmented policy can be unrolled into the same sequence of primitive actions and discounted rewards. But the probability of proposing the full low path is now q_o rather than q^M. This is the exponential temporal-abstraction gap shown in figure 1A.

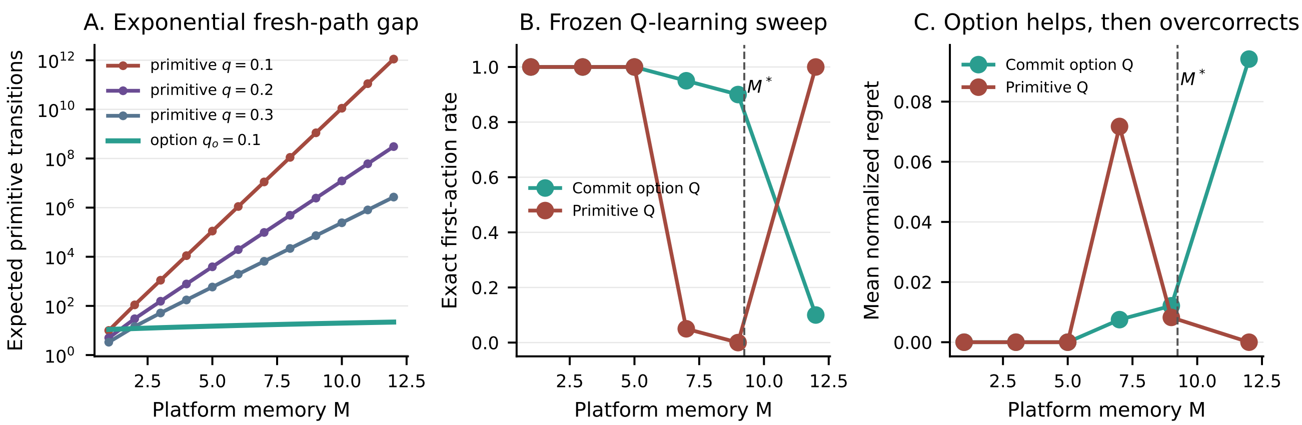

Figure 1: Theory and frozen experiment. A: after any readiness checkpoint, iid primitive exploration takes exponential additional time to produce a fresh length-M cut path; a commitment option is linear in M. B: exact first-action agreement over 20 seeds. C: mean normalized regret. The dashed rational boundary $M^* = 9.2402$ is computed from payoffs, not fitted to learner outcomes.

6 Experimental design

The experiment is a controlled finite-MDP mechanism test, not a reduced-form test on observed seller prices. Its high and low quotes, payoffs, and penalty are frozen before simulation and chosen to satisfy equation (1); they are not structural estimates from a live marketplace. Exact value iteration supplies the target action. Tabular Q-learning uses 300,000 primitive transitions, discount 0.95, and 20 prespecified seeds. The option learner receives the same primitive transition budget; an option consumes its full duration. A 10,000-transition frozen-policy evaluation reports first-action agreement and normalized regret. Paired seed bootstrap intervals describe simulation variation only.

We use a falsification ladder rather than interpreting one favorable run. E-SIM5 gives the primitive learner the complete Markov history. E-SIM6 adds only the path-equivalent option and sweeps $M \in \{1, 3, 5, 7, 9, 12\}$. E-SIM7 transports the intervention to four frozen price books. E-SIM8 varies learning rate and exploration decay in a 3×3 grid. E-SIM9 replaces one-step Q with an eight-step TD target while retaining the primitive action set. All gates, seeds, and result hashes were frozen before interpretation.

After freezing E-SIM6, we added a descriptive trace audit rather than modifying its confirmatory gate. It replays the exact 20 primitive seeds, asserts every terminal Q-table against the frozen artifact to tolerance 10^{-12}, records trailing-low-state visits after steps 100,000 and 200,000, and reconstructs the deterministic empirical transition table from visited pairs. Batch Bellman sweeps then test whether coverage is sufficient when update order is repaired. The checkpoints are diagnostics, not preregistered readiness estimates.

Cross-sectional simulation. SM3 holds total provider count fixed and evaluates theorem 6 on an integer grid for $\beta \in \{0, .25, .5, .75, 1\}$. This is a numerical theorem check. A separate paired bandit experiment preserves each learner's marginal signal sequence but changes common signal order, then measures action correlation and active share. Seed-clustered intervals are frozen before interpretation. This second stage is the transport test: signal coupling must move allocation, not merely the correlation statistic used to construct it.

Public-market property tests. The public panel is frozen at immutable revision `f4023bdad64b6ad468741a5a3b1f3afb7af40dd0`. It contains 3,182,780 endpoint observations over 3,274 five-minute snapshots from 7–22 July 2026. Free model routes are excluded. Active undercutters are classified before a temporal split by at least two price changes and a median quote below the author/menu benchmark. We estimate effective rank for nested active sets and block-bootstrap the log-rank slope over calendar blocks. We also reset active quotes to the benchmark and recompute inverse-price shadow shares. These objects do not enter the MDP payoffs, M, transition law, or result selection. A contemporaneous rank or score wedge cannot identify router memory.

The owned GLM-5.2 intervention compares default routing with an explicit price-sort request within complete randomized blocks. Only assignment-level aggregates are used here. The preregistered confirmatory gate requires 800 choices, 100 blocks, seven days, and 90% candidate coverage; current estimates are labeled accruing.

7 Results

Critical memory is visible in learning. At $M \in \{1, 3, 5\}$, primitive and option learners agree with exact control in 20/20 seeds. At $M = 7$, where exact optimization still cuts, primitive Q agrees in 1/20 seeds while option Q agrees in 19/20; 18/20 option seeds also satisfy the joint low-regret gate. At $M = 9$, the comparison is 0/20 versus 18/20. The transition is therefore near, but below, the independently derived rational boundary.

At $M = 7$, mean normalized regret is 0.07175 for primitive Q and 0.00750 for option Q. The paired difference is -0.06425 with 95% seed-bootstrap interval $[-0.07553, -0.04930]$. The sign remains negative in every one of nine learning-rate/exploration-decay cells; the strict composite severity gate passes in seven.

Early discovery is not the explanation. The trace audit reproduces all 20 frozen Q-tables exactly. Every primitive learner reaches the all-low state early: first-hit times range from step 7 to 901. Every seed also observes every one of the 256 state–action pairs at least nine times. Nevertheless, 19 finish with the wrong high initial action. Among those failures, none visits a state with five or more trailing low actions after step 100,000, and none returns to the all-low state. Thus neither first discovery nor aggregate support is the explanation. Ordered batch Bellman sweeps on each seed's same observed deterministic transitions recover the exact initial cut in 20/20 seeds. The observed mechanism is therefore *late credit starvation*: the data needed to solve the model were collected while exploration was broad, but online one-step updates did not revisit deep states after downstream values separated. This diagnostic motivates the stopping-time formulation of theorem 2; it does not estimate a unique readiness time.

The intervention is nonmonotone. At $M = 12$, exact optimization stays high. Primitive Q is correct in 20/20 seeds, whereas option Q is correct in only 2/20 and adds 0.0942 normalized regret. An interface that solves delayed credit can therefore reduce welfare when it creates commitment salience on the wrong side of the economic boundary. Mechanism evaluation must combine theorem 1 with theorem 2; the learning boundary alone is not a design objective.

The falsifications narrow the mechanism. Complete state observability does not repair primitive learning: it remains high in 19/20 seeds at $M = 7$. The severe-trap transport gate fails in two of four external price books, although all four option-effect signs align with their rational boundary. Finally, an ordinary eight-step TD target succeeds in 0/20 seeds and has regret difference $+0.0038$ relative to one-step Q, interval $[0, 0.0113]$. The supported result is option-specific temporal abstraction in the controlled MDP, not a claim that any longer return solves delayed credit.

The cross-sectional theorem transports algebraically, not behaviorally. For $\beta = 0, .25, .5, .75, 1$, the analytic exponents in equation (14) are $.5, .625, .75, .875, 1$. Integer optima across the registered n grid recover $.498, .620, .745, .876, 1.005$; the maximum error is 0.006. This verifies the discrete implementation, not the congestion loss. In the paired bandit intervention, the active-share effects are -0.00116, -0.00086, and -0.00038 for rank exponents $0, .5, 1$; all seed-clustered intervals include zero. Action correlation can move without a distinguishable allocation effect. This negative control rejects the claim that common signal order alone operationalizes theorem 6 in the declared learner.

8 Public-market property tests

The empirical exercise asks which proposed cross-sectional regimes remain compatible with the frozen data. It does not estimate M, provider learning algorithms, realized market share, or k_n^*.

GLM-5.2 has a large shadow-share surface but no sublinear-rank result. The frozen classifier selects Novita, StreamLake, and Inceptron: three active undercutters in a median menu of 25. Under the public inverse-price surface, their post-split shadow share is 27.61%; resetting only those quotes to the benchmark gives 8.99%, a passive-to-active transfer of 18.62 percentage points. This is a deterministic rule counterfactual. At a one-hour innovation horizon, the nested-set effective-rank slope is 0.967 with block-bootstrap 95% interval $[0.516, 1.219]$; at six and 24 hours the point estimates are 1.098 and 1.121. GLM-5.2 is therefore linear-compatible at every horizon, not evidence for $k^* = o(n)$.

Non-GLM repricing is sparse, not linear-density. Across 70 paid non-GLM markets, 90% have zero active undercutters and the median active density is zero. The cross-market log-count slope on menu size is 0.219 with model-bootstrap interval $[0.068, 0.378]$. Only `openai/gpt-oss-120b` supports a three-provider rank ladder; its strong training factor loses roughly half or more of its explained variance in the holdout. These finite markets do not support $k = \Omega(n)$.

The direct rule effect is promising but accruing. In 15 complete owned-request blocks, default routing selects the cheapest provider 0% of the time and explicit price sorting selects it 80% of the time; the block-bootstrap difference is $[60, 100]$ percentage points. The public price exponent fitted to 36 default choices is 1.735 with profile interval $[0.31, 3.16]$. These estimates show that the action interface can materially change allocation for one account. They do not pass the registered duration,

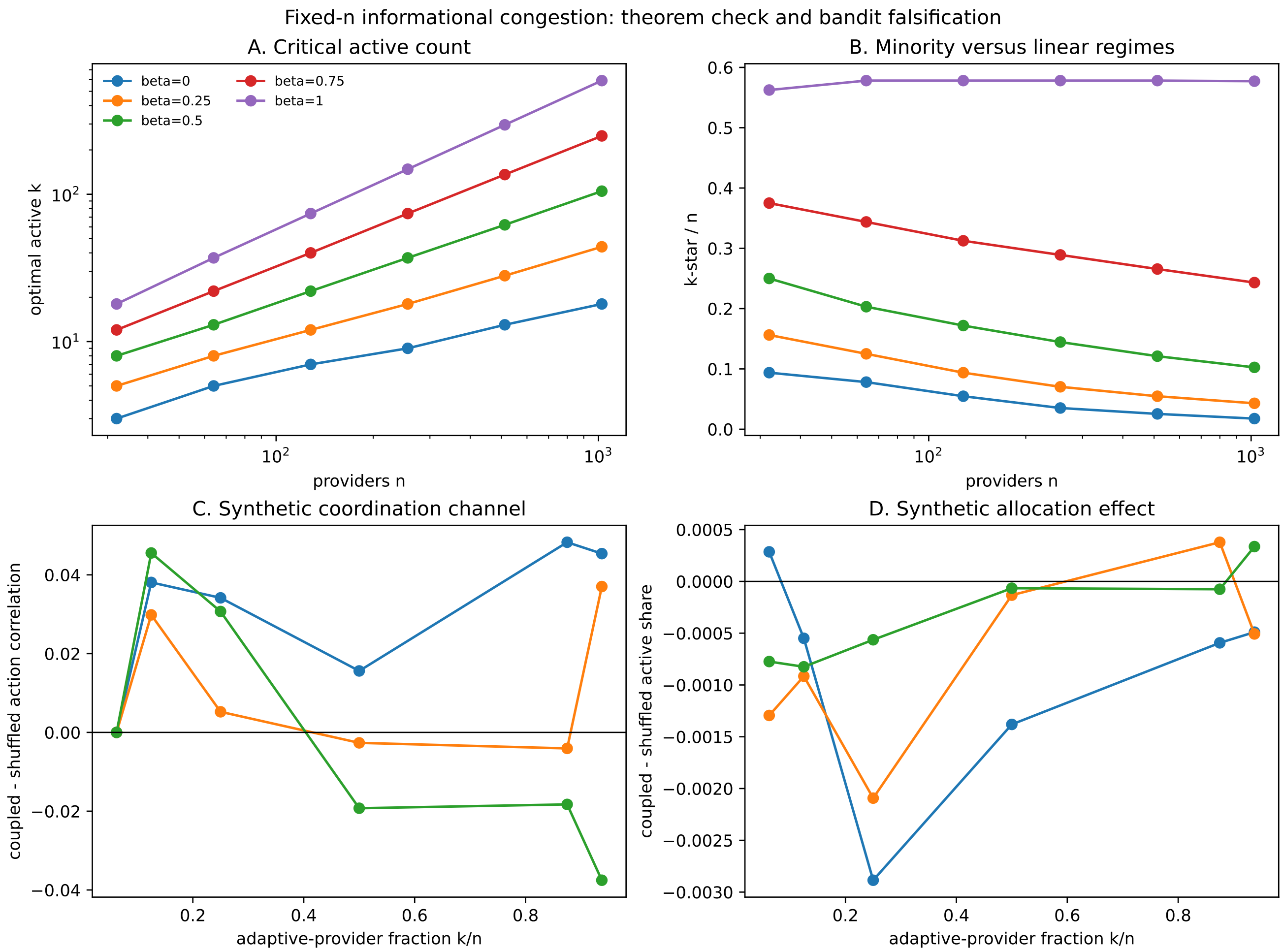

Figure 2: Fixed-n information-congestion check and behavioral falsification. Panels A–B recover the conditional optimizer in theorem 6. Panels C–D show that the marginal-preserving signal intervention can alter simulated action correlation while its active-share effect remains economically and statistically indistinguishable from zero. Theory recovery is not evidence that the declared bandit generates the loss.

sample-size, or coverage gates and do not reveal historical memory.

9 Implications for stateful platform design

The two boundaries separate three regimes.

1. If $M > M^*$, a persistent cut destroys provider value. High pricing is rational; exploration policy is secondary.
2. If $M \leq M^*$ but $M > M_c(T, q, \delta)$, cutting is rational but is unlikely to receive a fresh length-M support path within the late-learning horizon. A platform can sustain high learned prices without communication or supra-rational reward coupling.
3. If $M \leq \min\{M^*, M_c\}$, ordinary exploration can expose the low path; ordinary reward noise and unmodeled cross-agent feedback again become first-order.

This yields a design warning. Platform memory can look anti-gaming because it disciplines short-lived cuts, yet the same rule can suppress efficient persistent cuts by altering learnability. Conversely, exposing a commitment action can restore price competition below M^* but cause excessive cutting above it. A safer platform would expose the state and a cancellable multi-period quote schedule only when a deviation audit predicts that persistent cutting raises provider value and buyer surplus. The theorem supplies a testable screen, not a universally optimal policy.

When signal groups are observable, a direct exposure cap is preferable to manufacturing a long memory barrier. The router can bound aggregate attempt probability assigned to providers with correlated quote innovations, preserve a small exploration floor, and expose a cancellable multi-period action that lets learners reveal persistent intent. The cap targets cross-sectional congestion; the option targets temporal credit. Deploying either alone confounds the two problems.

Illustrative inference-routing application. The target score must also match the economic objective. A

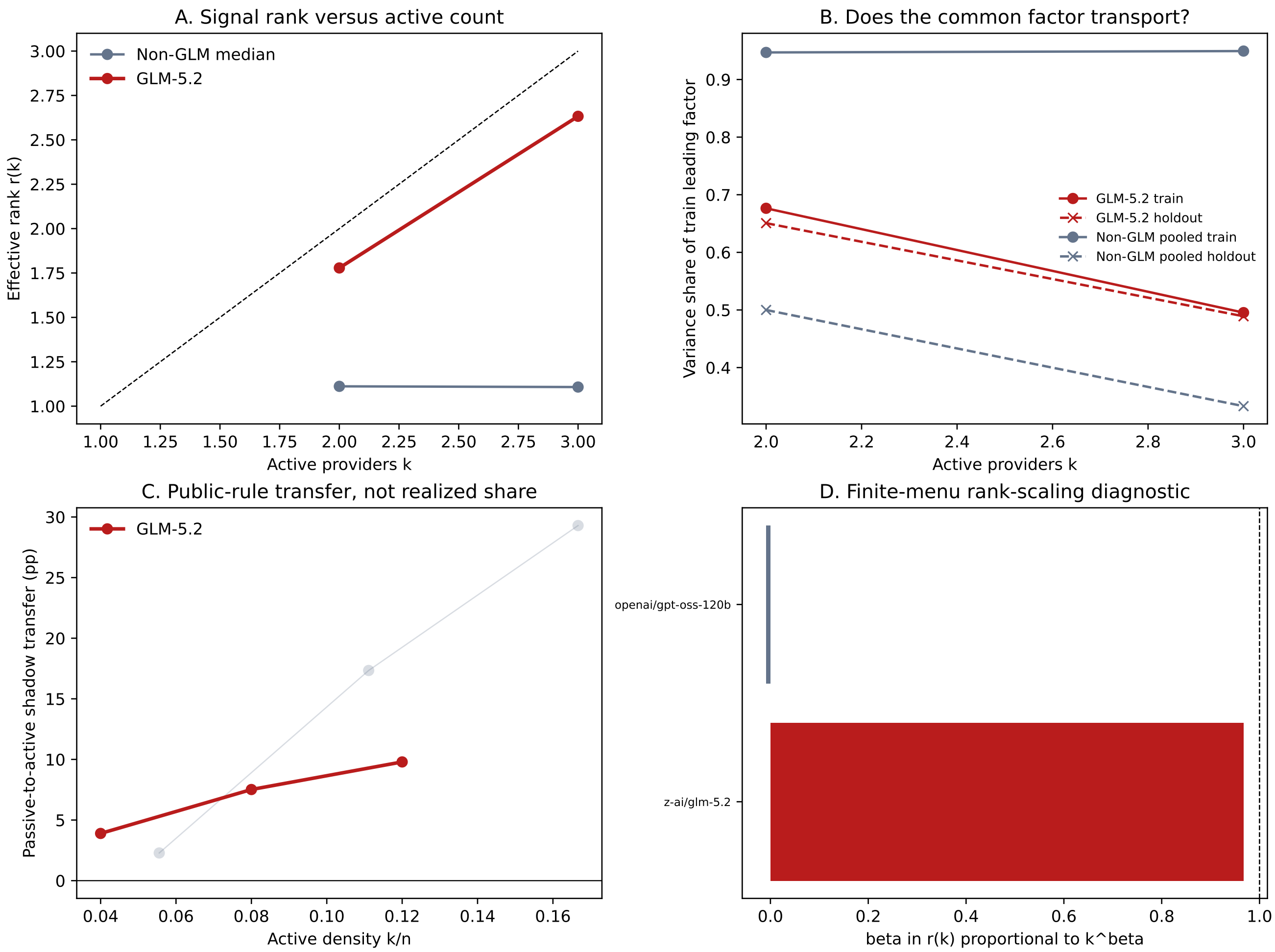

Figure 3: Frozen public-market property test. GLM-5.2 has a small active set and large public-rule shadow transfer, but its nested effective rank is linear-compatible. Non-GLM activity is sparse and its only low-rank ladder fails temporal transport. The slopes are finite-menu diagnostics, not asymptotic estimates or realized route shares.

user-cost router scores payment, latency, and failure; a welfare router substitutes resource cost for payment and values delivered quality; a revenue router may favor high spend under ad-valorem fees; and a quality-priority router maximizes verified fidelity subject to budget. These objectives generate different payoff triples and hence different M^*. The commitment option should be evaluated on a Pareto frontier over user cost, welfare bounds, router revenue, quality, completion, and provider exploitability rather than on price alone.

Provider technology shifts the same boundary. An owned or reserved-capacity provider pays high fixed cost but may rationally cut to maintain utilization; a spot-dependent provider has less sunk exposure but state-dependent scarcity cost; an anchor adopter may not update displayed price at all. A router that hides score memory can therefore trap low-short-run-cost providers while doing little to passive anchors. This is a theoretical comparative static, not an interpretation of the current provider panel. Practical implementation should expose the score state, offer cancellable multi-period quote commitments, use execution-contingent capacity rather than fundraising or provider identity as a proxy, and charge fees separately from selected spend.

Objective-conditioned equilibrium screen. A separate structural game declares ten provider technologies: an author, two anchor resellers, two reserved-capacity providers, three spot-dependent startups, and two custom-quality providers. It recomputes bounded cyclic best responses under 108 router rules. In simulation units, the inverse-square price-only baseline yields welfare 137.29 and router fee revenue 5.99. The welfare-selected rule yields welfare 208.44 and revenue 4.88, whereas the revenue-selected rule yields revenue 7.78, welfare 83.99, and negative user utility. Objective-selected rules have zero unilateral grid regret but retain profitable bounded pair deviations. Thus a learning intervention must be evaluated separately for welfare, router revenue, delivered quality, user utility, provider profit, and viability. The technologies are declared stress scenarios, not estimates for named providers.

10 Related work

Algorithmic-pricing research establishes that independent learners may support high prices and that results depend sharply on algorithms and timing (Calvano et al., 2020; Asker et al., 2022; Brown and MacKay, 2023). Hansen–Misra–Pai show how independent bandit experiments become strategically coupled through the aggregate price used to score arms (Hansen et al., 2021). Johnson–Rhodes–Wildenbeest study platform steering when sellers use pricing algorithms (Johnson et al., 2023). Our contribution is orthogonal: a platform's own-history rule creates a finite-time fresh-support barrier even when reward differences are otherwise easy to estimate.

Options and semi-Markov value preservation are classical (Sutton et al., 1999); delayed reward and sequence compression have recently been studied directly (Han et al., 2022; Ramesh et al., 2024). We add an economic source of the delay, an explicit critical-memory rate, a rational boundary with the opposite comparative-static force, and controlled evidence that the representation intervention helps only between those boundaries.

11 Limitations and claim boundary

The finite-time theorem assumes a post-stopping-time cut-probability cap and applies only to learners that require fresh downstream support. It does not characterize algorithms that plan over a known model, retain replay sufficient for exact offline backups, or deliberately schedule low runs. The diagnostic checkpoint is post hoc and does not identify when a Q-value becomes usable. The controlled experiment fixes rivals, uses two prices, and studies tabular Q. Twenty seeds identify Monte Carlo variability, not uncertainty in live-market parameters. Cross-profile severe-trap transport fails, and no paid probe randomizes a memory rule. The information-congestion result additionally assumes the form of the common-mode loss; effective rank alone cannot identify its coefficient or welfare meaning. SM3's behavioral stage is null. WF20 has only two GLM rank points, a short temporal panel, public shadow shares rather than realized flow, and a sparse non-GLM comparison. We therefore claim finite-time bounds, a conditional planner theorem, controlled mechanisms, and frozen property tests—not observed market share, a GLM/non-GLM asymptotic contrast, provider intent, tacit collusion, communication, or live-router causality.

12 Conclusion

Platform memory creates three separable objects: whether persistent cutting is valuable, whether a learner can obtain fresh support in finite time, and how many providers should respond to correlated information. The first is a rational boundary; the second is exponential in memory; the third depends on signal rank and the declared congestion loss. A path-equivalent option repairs late credit but overcorrects beyond the rational boundary. A covariance-aware cap targets cross-sectional exposure but requires realized-outcome calibration. The public market currently shows large mechanical share transfer and sharply sparse adaptation, yet it supports neither proposed asymptotic regime. The joint lesson is methodological: separate individual elasticity, finite-time learnability, and social information congestion, then demand a transport test at each link before calling a learning rule a market mechanism.

A Proofs

Proof of theorem 1. Once the history is all-low, L strictly dominates H by equation (1); choosing H also restarts the penalty. Before that state, any policy that eventually cuts is weakly improved to k high actions followed by low forever. Its value is

$$V_k = \frac{u_H(1-\gamma^k)}{1-\gamma} + \gamma^k V_{\text{cut}},$$
$$V_{\text{cut}} = \frac{(1-\gamma^M)u_{\theta L} + \gamma^M u_L}{1-\gamma}.$$

Because $V_k - u_H/(1-\gamma) = \gamma^k[V_{\text{cut}} - u_H/(1-\gamma)]$, either $k = 0$ or never cutting is optimal. Comparing these alternatives yields equation (2). □

Proof of theorem 2. Condition on $\mathcal{F}_\sigma$. For each candidate ending time $t > \sigma$, iterated conditioning and equation (3) bound the probability of the required post-σ all-low length-M window by q^M. A union bound over $(T-M+1)_+$ windows proves equation (4); inversion of its looser Tq^M form gives equation (5). Under iid Bernoulli exploration, let E_k be the expected remaining time after a current run of k lows. The recursions $E_M = 0$ and $E_k = 1 + qE_{k+1} + (1-q)E_0$ solve to $E_0 = (1-q^M)/[(1-q)q^M]$. □

Proof of theorem 4. Apply theorem 2 to each provider and truncate the resulting upper bound at one. Linearity of conditional expectation does not require cross-provider independence. Markov's inequality gives the tail bound. □

Proof of theorem 5. For inverse-power routing,

$$\frac{\partial s_i}{\partial \log p_i} = -as_i(1-s_i), \qquad \frac{\partial s_j}{\partial \log p_i} = as_is_j \quad (j \neq i).$$

Hence the ith Jacobian column has ℓ_1 norm $2as_i(1-s_i)$. The triangle inequality bounds the local total variation by $a\sum_i s_i(1-s_i)|d_i|I_i(T)$. Use $|d_i| \leq \Delta$, take conditional expectations, and apply theorem 4 provider by provider. □

Proof of theorem 6. Substituting equation (12) gives

$$V_n(k) = \frac{bk}{n} - \frac{ck^{2+\alpha}}{n^2 r_n^\alpha}.$$

Its derivative vanishes only at equation (13); the second derivative is strictly negative for $k > 0$. Strict concavity places the integer maximizer at an adjacent integer. Substituting $r_n \asymp n^\beta$ gives equation (14). $\square$

Proof of theorem 7. Under homogeneous (M, q), theorem 4 gives $\mathbb{E}[K_T] \leq nTq^M$ in the nontrivial region. Solving $nTq^M \leq k_n^*$ for M gives equation (15). Substitute $k_n^* = \Theta(n^\zeta)$ and collect logarithms. $\square$

Proof of theorem 8. The augmented action set contains every primitive action, so its optimal value is weakly larger. Replace each option in any augmented policy by its defining $M+1$ primitive low actions. The unrolled policy produces identical states, durations, and discounted rewards, so its value is identical and the augmented optimum cannot be larger. Before the first option, the number of one-step decisions is geometric with mean $(1-q_o)/q_o$; adding the final $M+1$ transitions gives $M + q_o^{-1}$. $\square$

B Reproducibility

The theory functions and unit tests are in `src/orcap/market_env/critical_memory.py` and `tests/market_env/test_critical_memory.py`. The figure is generated directly from the frozen E-SIM6 parquet. Clean bundles are E-SIM5 `3e5a55405e`, E-SIM6 `93265b9b03`, E-SIM7 `bd74ab2eb7`, E-SIM8 `4d84b9b3a2`, and E-SIM9 `179eca0f9d`. Manifests pin source commits, seeds, result hashes, and input-artifact hashes. Exact Bellman residuals, path equivalence, paper numbers, and the new finite-time identities are enforced by tests. The post-hoc trace artifact `credit-propagation-diagnostic.json` is generated by the adjacent replay script. It is admitted only after all 20 Q-tables match the frozen E-SIM6 artifact; separate tests reconcile seed-level actions, early hits, and late depth counts. SM3 is generated by `src/orcap/analysis/sm3_informational_congestion.py`; its unit tests check effective-rank algebra, the continuous optimizer, integer adjacency, fixed-n scaling, paired signal marginals, and seed-clustered intervals. WF20 is generated from immutable public-data revision `f4023bdad64b6ad468741a5a3b1f3afb7af40dd0`; its outputs include the model census, nested rank curves, block bootstrap, and benchmark-reset shadow shares. A clean public integrity run completes 29 registered modules with zero failures and does not query prospective owned outcomes. SM4 separately checks welfare accounting, fixed-provider menus, capacity redistribution, bounded best responses, objective selection, and JSON serialization. None of WF20 or SM4 enters this paper's MDP payoffs, memory length, transition law, or confirmatory gates. The detailed boundary is in the dated information-congestion evidence audit adjacent to the venue folders.

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